Minghua Liu

+86-166-9158-2338 | liuminghua24@mails.ucas.ac.cn | Website | GitHub
Beijing, China

Education

- **Institute of Software, Chinese Academy of Sciences** *2024.09 - 2027.07(Expected)*
  M.S. in Artificial Intelligence Beijing, China
  - **Advisor: Lijun Zhang**
  - **Research Interest**: Trustworthy LLMs, including uncertainty quantification, explanation, and their applications
  - GPA: 3.93/4.00
- **Beijing Information Science & Technology University** *2020.09 - 2024.06*
  B.S. in Information and Computing Science (Computational Mathematics) Beijing, China
  - GPA: 4.12/5.00

Publications and Manuscripts

[1]] Yanni Dong, **Minghua Liu**, Meilin Zhu, Xiaowei Huang, and Lijun Zhang. **Beyond Semantic Equivalence: Logical Graphs for LLM Uncertainty Quantification**. arXiv:2607.16868, 2026. Manuscript under review; not an accepted conference paper.

[2]] **Minghua Liu**, Lele Wang, Heng Ge, et al. **Research and Implementation of an Academic Mutual-Assistance Platform Based on a WeChat Mini Program** (English title translated). *Microprocessors*, 44(5):21–26, 2023. Published in Chinese.

Research Projects

- **Logical Graphs for LLM Uncertainty Quantification** *[Sep 2025] - Present*
  Tools: Python, Llama, Falcon, Qwen, Mistral [GitHub]
  - Constructed logical graphs on top of semantic clustering to address the over-detection of hallucinations in LLM outputs.
  - Implemented probability propagation within the graph and introduced a logical contradiction coefficient to effectively recalibrate uncertainty.
  - Evaluated on four QA benchmarks; the preprint reports gains of up to **7.1 percentage points in AUROC** and **3.5 points in AUARC** over semantic entropy on questions with logically structured answer sets.
- **Probing Transformer Factual Memory via Sparse FFN Perturbation** *[Mar 2026] - [Jun 2026]*
  Tools: Python, PyTorch, Transformers, Llama [GitHub]
  - Applied sparse Gaussian perturbations to intermediate Transformer FFN activations while keeping generated tokens fixed, then measured the resulting probability variation as an internal uncertainty signal.
  - Evaluated the manuscript method on factual-recall and reasoning tasks; the reported Llama-3.2-1B Math500 AUROC is **82.37%**, versus **72.51%** for the strongest listed baseline. This work is not an accepted publication.
- **Deep Learning for Protein Phosphorylation Site Prediction** *[2022] - [2024]*
  Tools: ESM-2, FCNN, PTMiner, Python [Link]
  - Built an ESM-2 representation and fully connected neural network (FCNN) pipeline to predict phosphorylation sites from protein-sequence windows.
  - Used sequence-based probabilities alongside PTMiner mass-spectrometry evidence to propose candidate sites when spectra could not localize a modification confidently.
  - The undergraduate thesis received a **Beijing Outstanding Undergraduate Thesis** award.

Selected Open-Source Work

- **UCRA: Uncertainty-Guided Cross-Source Response Acquisition** *2026*
  Original research prototype; Python
  - Released a testable policy that uses semantic uncertainty to decide when to query another model and fuses answers with source-aware reliability and dependence weights.
- **lark-codex-bridge** *2026*
  Open-source adaptation of an existing bridge architecture; TypeScript
  - Adapted a Feishu/Lark agent bridge for the local Codex CLI, including per-chat sessions, streamed replies, workspace controls, file handling, and access settings.
- **DataDoctor (ForgeAI Robotics)** *2026*
  Contribution to an open-source LeRobot data-quality toolkit; Python
  - Contributed a full-video audit that decodes frames, cross-checks visual motion with robot state, and reports anomalous frame/time intervals; the toolkit builds on upstream *lerobot-doctor*.

Experience

- **ByteDance [🌐]** *May 2026 - Aug 2026*
  AI R&D Intern, Seed Pretraining Data Engineering Beijing, China
  - **Tech Stack:** Python, Spark, MoE, LLM, Codex, Agent, OCR.
  - Spearheaded specific field annotation and completion for over **800 million** data samples within the Seed pretraining data pipeline, establishing robust large-scale data processing and quality control workflows.
  - Fine-tuned a **1.1B MoE (Mixture of Experts) model** for document classification and quality scoring, leveraging the model's reasoning capabilities for automated data quality identification and sample filtering.
  - Constructed an automated evaluation pipeline for PDF understanding tasks, integrating traditional OCR with LLMs to conduct multi-dimensional quality assessments and provide automated feedback on document parsing.
  - Developed a Codex-based automated workflow Agent to orchestrate data processing, task execution, and result feedback, pioneering LLM Agent applications in data engineering automation.
- **ByteDance [🌐]** *Sep 2023 - Dec 2023*
  NLP Intern Beijing, China
  - **Tech Stack:** Python, LLM Agent, Prompt Engineering, Code Execution.
  - Designed and implemented an LLM Agent for tabular calculation tasks, building an end-to-end automated processing pipeline: **Table Parsing** → **LLM Reasoning** → **Python Code Gen** → **Code Execution** → **Result Validation** → **LLM Polishing**.
  - Decoupled deterministic calculations from LLM reasoning by utilizing **Tool Calling** to invoke a Python execution environment, significantly reducing computational errors and hallucinations caused by direct numerical reasoning.
  - Engineered Agent workflows, complex prompts, and tool-calling logic to achieve a fully automated task chain, seamlessly transforming structured data inputs into accurate natural language outputs.

Skills

- **AI & Agent:** LLM evaluation, uncertainty quantification, tool-calling workflows, Codex CLI
- **Programming Languages:** Python, C/C++, TypeScript
- **Data Science & Machine Learning:** PyTorch, scikit-learn, Transformers, Spark

Honors and Awards

- **Beijing Outstanding Undergraduate Thesis** *Jun 2024*
  Beijing Municipal Education Commission / BISTU
  - Recognized the ESM-2 + FCNN phosphorylation-site prediction and PTMiner-assisted localization thesis.
- **Meritorious Winner** *Apr 2022*
  Interdisciplinary Contest in Modeling (ICM/MCM)
  - Developed a mathematical model and technical report under the competition's time limit.
- **First Prize (National & Provincial Level)** *2021 & 2022*
  Chinese & Beijing College Students Mathematics Competitions
  - Received first-prize awards at the national and Beijing municipal levels.